

● Review

FOR WHOM THE BUBBLE GROWS: PHYSICAL PRINCIPLES OF BUBBLE NUCLEATION AND DYNAMICS IN HISTOTRIPSY ULTRASOUND THERAPY

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Abstract—Histotripsy is a focused ultrasound therapy for non-invasive tissue ablation. Unlike thermally ablative forms of therapeutic ultrasound, histotripsy relies on the mechanical action of bubble clouds for tissue destruction. Although acoustic bubble activity is often characterized as chaotic, the short-duration histotripsy pulses produce a unique and consistent type of cavitation for tissue destruction. In this review, the action of histotripsy-induced bubbles is discussed. Sources of bubble nuclei are reviewed, and bubble activity over the course of single and multiple pulses is outlined. Recent innovations in terms of novel acoustic excitations, exogenous nuclei for targeted ablation and histotripsy-enhanced drug delivery and image guidance metrics are discussed. Finally, gaps in knowledge of the histotripsy process are highlighted, along with suggested means to expedite widespread clinical utilization of histotripsy. (E-mail: baderk@uchicago.edu) © 2019 The Author(s). Published by Elsevier Inc. on behalf of World Federation for Ultrasound in Medicine & Biology. This is an open access article under the CC BY-NC-ND license. (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Key Words: Histotripsy, Focused ultrasound, Cavitation, Ablation, Bubbles.

INTRODUCTION

One of the primary goals of new medical technologies is the reduction of morbidity and complications in the course of treating a disease (Harrell and Heniford 2005; Jaffray 2005). Surgery is not well tolerated in elderly or comorbid patients (Turrentine et al. 2006). The development of new surgical techniques or instrumentation focuses on reducing procedural complications, incisions, scarring and recovery time (Jaffray 2005; Tejjwani et al. 2017). Minimally invasive techniques such as laparoscopy and endoscopy achieve these goals, and are now standard procedure over open surgeries for many indications.

One of the most common minimally invasive procedures is ablation. The clinical endpoint of ablation is to render the target tissue biologically non-viable. Minimally invasive ablation can be achieved by several modalities, including radiofrequency alternating currents (Ni et al. 2005), microwaves (Lubner et al. 2010), laser energy (Gilling 2007; Oto et al. 2013) and freezing (Tatli

et al. 2009). Proton beam therapy (Levin et al. 2005), X-ray or gamma radiation (Lindquist 1995) and focused ultrasound (ter Haar and Coussios 2007) are non-invasive options for ablation. Thermal ablation by high-intensity focused ultrasound (HIFU) is a promising technology for precise reduction of tumors or benign growths (Chaussy and Thüroff 2014; Kennedy et al. 2004; Xu et al. 2004). Although the ablative efficacy of thermal HIFU has been reported for select pathologies, it is limited by long treatment times (Lu et al. 2003) and requires magnetic resonance imaging (MRI) thermometry or quantitative B-mode imaging (Tavakkoli and Sanghvi 2011) for real-time feedback of treatment. Furthermore, thermal ablation of highly perfused tissue can be imprecise, inconsistent or ineffective (Curley 2001).

An alternative to thermal ablation is mechanical disruption of tissue, such as thrombectomy (Stähr et al. 1999) or transurethral prostatectomy (Mebust et al. 1989). Histotripsy is a form of therapeutic ultrasound that produces mechanical disintegration of tissues that can be utilized to complement current ablative techniques or as a stand-alone modality (Parsons et al. 2006; Roberts et al. 2006; Xu et al. 2004). The delivery of high-amplitude, short-duration histotripsy pulses

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repeatedly strains tissue at the subcellular level (Parsons *et al.* 2006; Roberts *et al.* 2006; Vlaisavljevich *et al.* 2016d). Through application of a sufficient number of pulses, a target tissue structure can be homogenized precisely within the focal region. Within the treatment zone, cells are disintegrated into their subcellular constituents and the extracellular matrix is fragmented. Tissue outside the focal region remains intact (Fig. 1). Unlike thermal HIFU, the efficacy of histotripsy is not suppressed in highly vascularized tissue. Furthermore, both therapy delivery and the resulting tissue disruption can be monitored in real time with either ultrasound or MRI. The application of histotripsy to treat pathologic tissue is predominately in the pre-clinical phase. However, histotripsy has the potential to be utilized in disease pathologies that would otherwise require surgical intervention (Bader *et al.* 2018; Duryea *et al.* 2011; Hall *et al.* 2007b; Hempel *et al.* 2011; Khokhlova *et al.* 2014; Kim *et al.* 2013, 2014a; Maxwell *et al.* 2011a; Owens *et al.* 2010; Schade *et al.* 2015; Styn *et al.* 2010; Vlaisavljevich *et al.* 2013b, 2016c; Zhang *et al.* 2017). Pre-clinical studies have investigated histotripsy for the treatment of thrombus obstruction (Bader *et al.* 2016a; Maxwell *et al.* 2011a;

Sukovich *et al.* 2017; Zhang *et al.* 2017), fetal septal defects (Xu *et al.* 2010), kidney stones (Duryea *et al.* 2014), hepatocellular carcinoma (Khokhlova *et al.* 2014; Vlaisavljevich *et al.* 2013c, 2017b), renal carcinoma (Roberts *et al.* 2006), transcranial ablation (Sukovich *et al.* 2016) and biofilms (Bigelow *et al.* 2018), amongst others (Khokhlova *et al.* 2015). Furthermore, a clinical trial was recently completed whose primary outcome was to assess the safety of histotripsy technology to treat benign prostatic hyperplasia (Schuster *et al.* 2018).

Tissue ablation does not result directly from the ultrasound pulse, but *via* the formation of acoustic cavitation or boiling bubbles. Cavitation has been detected previously with therapeutic and diagnostic devices in *in vitro* (Atchley *et al.* 1988; Fowlkes and Crum 1988; Roy *et al.* 1990) and *in vivo* (Holland *et al.* 1996; Hynynen 1991; Sapozhnikov *et al.* 2001; ter Haar and Daniels 1981; ter Haar *et al.* 1986) studies. Bubble activity can enhance the therapeutic efficacy of several different clinical applications (Bader *et al.* 2015a; Coussios and Roy 2008). However, inciting the intended bubble activity is difficult because of the episodic nature of cavitation (Coleman *et al.* 1996; Haworth *et al.* 2015; Holt and Roy

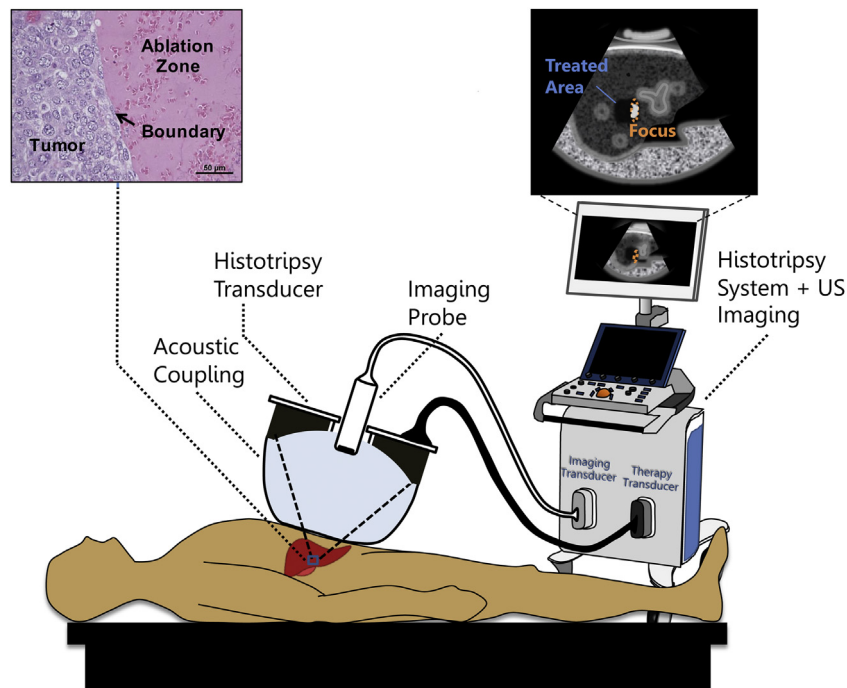


Fig. 1. Conceptual illustration of a non-invasive histotripsy procedure. A focused ultrasound transducer is coupled to the patient through a confined water bolus attached to the transducer face. The transducer also contains an ultrasound (US) imaging probe for targeting and guidance. Both transducers are controlled by a combined imaging/therapy system. The transducer focus is positioned within the target tissue using US imaging guidance. When therapy is administered, bubbles appear on the US image as a hyper-echoic region confined to the focus. Over a short time, the tissue is disintegrated into subcellular debris with a precise boundary. Once the tissue is ablated, it appears on imaging as a hypo-echoic area indicating to the operator that treatment is complete.

2001; Jensen et al. 2012; Sapozhnikov et al. 2001; Zhong et al. 1997). The stochastic behavior of cavitation can be attributed to the fact that separate aspects of bubble behavior (e.g., non-linear oscillations, jetting, inertial collapse) are dictated by separate acoustic pressure thresholds (Apfel 1981; Bader and Holland 2012; Church 1988). That is, for the same set of initial cavitation nuclei, different types of bubble behavior can be generated depending on the insonation conditions. Histotripsy overcomes the stochastic nature of cavitation by operating with sufficient tension to *always* induce massive expansion of bubble nuclei, thereby instigating only a very specific and predictable type of bubble activity.

Three focused ultrasound insonation schemes are utilized in histotripsy to produce repeatable bubble activity (Fig. 2), summarized here:

- *Intrinsic-threshold histotripsy* delivers pulses with a single tensile phase (0.33–10 μ s in duration) sufficient to cause a cluster of bubble nuclei intrinsic to the target medium to undergo inertial cavitation.
- *Shock-scattering histotripsy* applies pulses 3–20 cycles in duration. The amplitude of the tensile phases is sufficient to cause individual nuclei to undergo

inertial cavitation within the focal zone over the duration of the pulse. These bubbles scatter the incident shock waves geometrically, causing a dense bubble cloud to form in regions of constructive interference between the incident and scattered waves that exceeds the threshold for intrinsic nucleation.

- *Boiling histotripsy* employs pulses roughly 1–20 ms in duration. Absorption of the shocked pulse rapidly heats the medium, thereby reducing the threshold for intrinsic nuclei. Once this intrinsic threshold coincides with the peak negative pressure of the incident pulse, boiling bubbles form in the focus.

One of the key goals of this article is to explain how the different modes of histotripsy can be described in the context of a bubble population or threshold that appears to be intrinsic to a given medium. This review discusses theoretical and experimental aspects of histotripsy-generated cavitation and boiling, including pathways that reliably achieve dense bubble clusters necessary for tissue disintegration. The topics of bubble nucleation, cavitation and boiling by ultrasound have a vast history and body of literature. Discussion is necessarily limited to descriptions of those areas that achieve the specific effects described

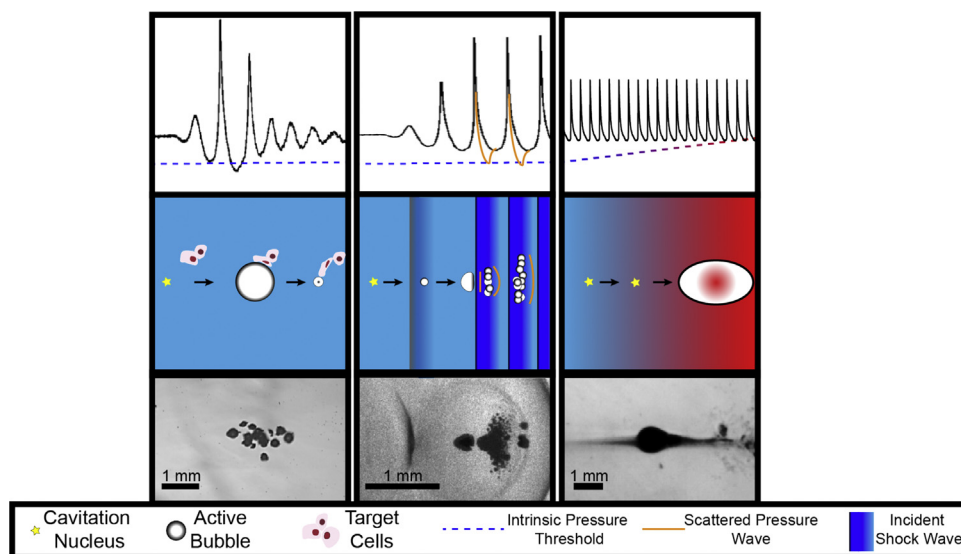


Fig. 2. Summary of histotripsy-induced cavitation dynamics. For all forms of histotripsy, nanoscale nuclei intrinsic to the medium are present in the tissue. Left column: For intrinsic threshold insonations, the cavitation nucleus is activated with a single-cycle pulse with tension below the intrinsic medium threshold (*left arrow*, middle row). The expanded bubble then undergoes an inertial collapse under ambient pressure (*right arrow*, middle row). Middle column: In shock-scattering histotripsy excitations, the activated nucleus grows slowly over the course of several cycles (*left arrow*, middle row) and deforms because of the incident shock waves (*right arrow*, middle row). Additional bubbles form spatially and temporally in regions of constructive interference between the incident wave and waves scattered by the deformed bubble. Right column: In boiling histotripsy, shock-enhanced heating alters the cavitation nucleus (*left arrow*, middle row) to reduce the requisite tension for bubble formation (*right arrow*, middle row). For all forms of histotripsy, the expansion and contraction of the bubble impart lethal strain on the cellular and extracellular components of the tissue in close proximity to the bubble (depicted in left column, middle row only). Representative frames from high-speed videography of histotripsy-generated bubbles are in the bottom row. Note: Bubble sizes and nuclei in the second row are not to scale.

in histotripsy. Whenever possible, discussion is tied to relevant historical precursors and literature that forms a foundation for the observations of bubbles in histotripsy. To gauge bubble activity, Apfel (1981) formulated three golden rules: (i) Know thy sound field, (ii) know thy liquid and (iii) know when something happens. Our understanding of histotripsy-induced bubble activity will be viewed through this framework.

KNOW THY SOUND FIELD

Although ultrasound is known most ubiquitously as a diagnostic imaging modality, its therapeutic use dates back a century. Early studies on ultrasound-induced bio-effects were conducted after World War I, predominately continuing wartime observations of sonar-induced fish death (Harvey and Loomis 1928; Langevin 1920) or non-invasive heating (Freundlich *et al.* 1932). Therapeutic ultrasound came into earnest with William and Frank Fry at the University of Illinois (O'Brien and Dunn 2015). Their preliminary *in vivo* studies with focused sources (Fry 1952; Fry *et al.* 1950, 1970) culminated in the treatment of 88 patients with Parkinson's disease (Fry and Fry 1960) and cemented the use of ultrasound as a therapeutic modality.

Linear propagation in focused ultrasound

The diffraction field of a focused source provides a high intensity within the target volume, but low intensity in the intervening tissues. Ultrasound is commonly generated in histotripsy by piezoelectric transducers operating at frequencies near 1 MHz, with focusing produced by acoustic lenses or curvature of the piezoelectric material itself (Kim *et al.* 2014b). Spherically focused sources produce a focal region of high pressure amplitude, where the ratio of amplitudes between the focal peak and transducer surface is defined by the focal gain, G . For most histotripsy applications, large-aperture transducers with short focal distances are used to ensure G is greater than ~ 30 (Hall *et al.* 2009; Maxwell *et al.* 2009, 2017; Xu *et al.* 2004). The focus forms as a prolate ellipsoidal volume, with typical -6 -dB dimensions of ~ 10 mm along the acoustic axis and 1- to 2-mm width lateral to the acoustic axis in a linear (low-amplitude) regime (Rosnitskiy *et al.* 2017).

Non-linear propagation in focused ultrasound

In addition to the effects of linear propagation and focusing, the high-intensity output of a therapeutic source will induce non-linear distortion of the pressure waveform (Hallaj and Cleveland 1999). Distortion of the sinusoidal pressure waveform is due to a local variation in the pressure-dependent sound speed, $c(p_a)$, which can be expressed to first order as (Hamilton and Blackstock 1998)

$$c(p_a) = c_0 + \frac{\beta p_a(t)}{\rho_0 c_0} \quad (1)$$

where β is the coefficient of non-linearity of the medium, c_0 is the ambient medium sound speed, ρ_0 is the ambient medium density and p_a is the acoustic pressure at time t . The transition of energy from the fundamental frequency into higher harmonics results in a steepening of the waveform (see Fig. 3). With sufficient amplitude and propagation distance, any pressure pulse will be distorted into a sawtooth waveform (Pierce 1989). For a focused source, the combined effects of diffraction and non-linearity cause waveform distortion that leads to a shock wave (Fig. 3). The resultant pressure waveform in the focus is asymmetric, with a larger-amplitude and shorter-duration positive pressure in comparison to the negative pressure (Bessonova *et al.* 2009). Shock waves can be generated for all forms of histotripsy therapy, and

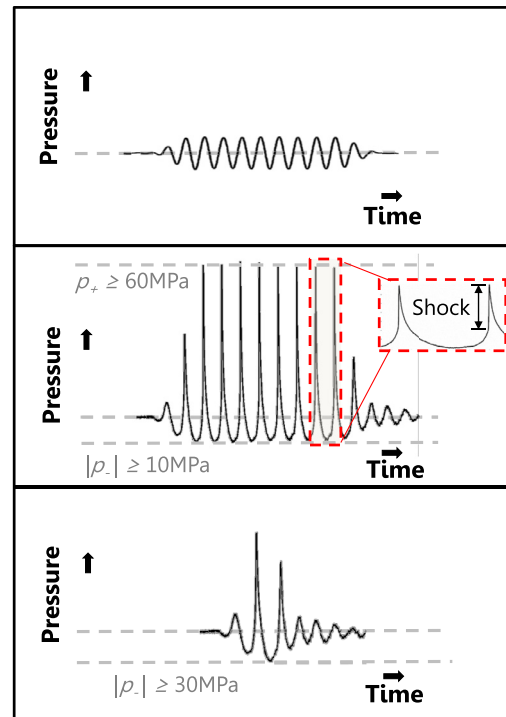


Fig. 3. Focal pressure waveforms generated by a source in the linear (top panel) and non-linear (bottom panel) regimes with a 10-cycle pulse. A short pulse, such as that used in intrinsic-threshold histotripsy, is illustrated in the middle panel. Note the highly asymmetric waveform in the middle and bottom panels, typical for focused sources due to the combined effects of diffraction and non-linear propagation.

are essential to generate the bubble activity necessary for ablation in shock-scattering and boiling histotripsy.

The degree of non-linear distortion in the pressure waveforms additionally depends on the transducer geometry and frequency. The most efficient means of generating large tension within the focus are sources with high focal gain or low f -number, defined for a spherically focused source as the ratio of focal distance to aperture diameter. As non-linear effects accumulate with propagation distance (Hamilton and Blackstock 1998), the pressure within the focus of low f -number sources (~ 0.5 – 0.7) consequently remains in a near-linear regime when generating cavitation (Vlaisavljevich et al. 2017a). As the f -number of the source increases, a non-linear wave will form, as depicted in Figure 3. Sources with f -numbers between 0.7 and 1 still have sufficient focal gain to promote cavitation activity via shocked excitations with fundamental frequencies generally less than 1.5 MHz. If the transducer is further unfocused (f -number > 1), robust shock waves are formed at relatively low-pressure amplitudes. The pressure waveform can reach a condition of acoustic saturation with a characteristic sawtooth waveform. At the acoustic saturation limit, the focal pressure does not increase with higher transducer output as the additional energy is dissipated in the shocks. Acoustic saturation occurs at pressures where bubble nucleation is stochastic for transducers with higher f -numbers.

KNOW THY LIQUID

The second of Apfel's golden rules denotes knowledge of cavitation nuclei within the medium. In this section, an overview of classical and heterogeneous nucleation theory in water is presented. This is followed by a discussion of potential nuclei found *in vivo* and a description of the influence of histotripsy insonation parameters on the cavitation threshold. A summary of the threshold for histotripsy-induced cavitation is also provided.

Bubble nucleation in water

Water mimics many acoustic properties of soft tissues. For this reason, the study of bubble formation in water has been an important focus in understanding histotripsy-induced cavitation. Flynn (1964) considered bubble nucleation to be a form of boiling, transitioning a liquid-based medium to a vaporous or gaseous state. The likelihood of bubble nucleation depends strongly on the quality of the water. In the case of pure liquids without exogenous nuclei, bubbles must be created through a first-order phase transition as described by classical nucleation theory (CNT). The relationship for the energy of a bubble to form W is the sum of the energy to displace the surrounding liquid volume, the generation of

vapor and the energy to create the bubble surface (Church 2002; Fisher 1948):

$$W = 4\pi R^2\sigma + \frac{4}{3}\pi R^3(p_0 + p_a - p_v) \quad (2)$$

Here R is the radius of the cavitation nucleus, σ is the medium surface tension, p_0 is the ambient pressure and p_v is the vapor pressure. Thermal fluctuations are the most ubiquitous source of energy in fluids. According to CNT, these fluctuations will produce bubbles ~ 1 Å in diameter spontaneously. The lifetime of these spontaneous nuclei is estimated to be $\sim 10^{-18}$ s as the Laplace pressure, given by $2\sigma/R$, causes the bubble to dissolve and the vapor to re-condense within the fluid (Yang and Prosperetti 2008).

The nucleation rate described by CNT is dictated by Maxwell–Boltzmann statistics (Church 2002). Thus, a pure liquid whose only source of nuclei is thermal fluctuations has no tension level at which there is a 100% certainty in cavitation. To allow direct comparison to the discussion in this article and other studies (Arvengas et al. 2011b; Herbert et al. 2006; Maxwell et al. 2013; Vlaisavljevich et al. 2015b), we define the cavitation threshold as peak negative pressure for which the probability of nucleating one or more bubbles during a pulse exceeds 50%. For histotripsy, this definition of the cavitation threshold is a useful measure because it indicates the pressure at which cavitation becomes likely to occur each pulse. The probability of producing cavitation can be altered by multiple factors. Applying tension over a larger volume of fluid or over a longer duration will result in an increased likelihood of cavitation (Church 2002). Similarly, heating or cooling the medium will alter the cavitation threshold, as indicated by the temperature-dependent vapor pressure and surface tension in eqn (2).

Efforts have been made to characterize the tensile strength of water (Briggs 1950; Couzens and Trevena 2002; Herbert et al. 2006; Zheng et al. 1991). One challenge in such a measurement is achieving water purity, as exogenous nuclei are not described by CNT. Measurement of the threshold tension in water predicted by CNT, 140 MPa, has only been achieved in quartz inclusion samples (Zheng et al. 1991) or within very short time frames (Stan et al. 2016). Efforts to replicate similar tensile values in water have yet to be achieved with other experimental designs. Picosecond X-ray (Stan et al. 2016) or nanosecond laser (Li et al. 2015) sources have generated cavitation at tensions of 100 and 60 MPa, respectively. Acoustic methods apply tension time scales on the order of 0.1 μ s to seconds, and an upper limit of 30-MPa tension has been measured before cavitation (Bader et al. 2012b; Briggs 1950; Herbert et al. 2006; Maxwell et al. 2013; Vlaisavljevich

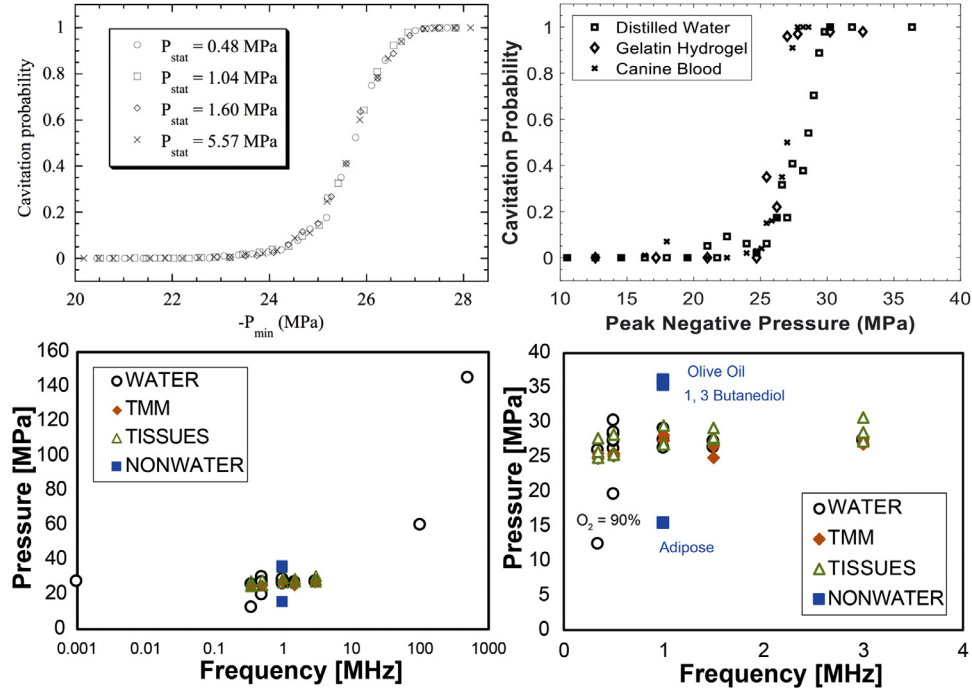


Fig. 4. Cavitation probability versus pulse peak negative pressure measured by Herbert *et al.* (2006) in water under different hydrostatic pressures (top left) and in several media by Maxwell *et al.* (2013) at 1 MHz (top right). These and other measurements suggest that the probability of histotripsy-induced cavitation is >0.5 for peak negative pressures >26 – 30 MPa. Bottom left: Pressure (in MPa) needed to achieve a 50% cavitation probability or the minimum pressure required to initiate cavitation detectable *via* acoustic backscatter ($P_{\min}$) for different materials (Maxwell *et al.* 2013; Vlaisavljevich *et al.* 2015b). Bottom right: The same plot focused on frequencies of interest for most histotripsy applications. TMM = tissue-mimicking material. Top left panel is reprinted from *Physical Review E*, vol. 74, Herbert *et al.*, Cavitation pressure in water, p. 041603 1–2. Copyright (2006), with permission from the American Physical Society.

et al. 2015b) (Fig. 4). Beyond 30-MPa tension, the probability of cavitation occurring is very close to 100% for such acoustic sources. Several factors may contribute to the discrepancy between pico- and micro-second nucleation thresholds. Chemical relaxation phenomenon may cause changes in water properties at different time scales (Pierce 1989). The formulation of bubble surface tension in bulk fluids is not well characterized at the molecular level, requiring simulations to refine the understanding of gas/water interfaces (Bruot and Caupin 2016; Bykov and Zeng 2001). Another possibility is the presence of liquid impurities that generate nuclei larger than the spontaneous nuclei associated with thermal fluctuations (Crum 1979; Fowlkes and Crum 1988; Harvey *et al.* 1944).

Sources of cavitation nuclei

When cavitation nuclei are present within the medium, the pressure necessary to cause explosive bubble growth can be predicted. The growth of an existing bubble nucleus in response to quasi-static changes in fluid tension can be described by the Blake threshold, p_l (Blake 1949):

$$p_l = p_0 - p_B = p_v - \frac{4\sigma}{3} \sqrt{\frac{2\sigma}{3(p_0 + \frac{2\sigma}{R_0} - p_v)}} \quad (3)$$

When the Blake threshold is exceeded (Fig. 5), the bubble will grow unbounded until the tension is removed or buoyancy causes the bubble to escape the liquid space. Although an acoustic wave is a dynamic pressure fluctuation, the measured threshold for histotripsy excitations (~ 30 MPa) corresponds to nuclei on the order of a few nanometers in diameter (Fig. 5). The resonant angular frequency of the nucleus ω_0 can be written in terms of the properties of the medium (Yang and Church 2005)

$$\omega_0 = \sqrt{\left\{ 3\kappa \left(p_0 + \frac{2\sigma}{R_0} - p_v \right) - \frac{2\sigma}{R_0} + p_v - \frac{4\mu^2}{\rho R_0^2} + 4G \right\} / (\rho R_0^2)} \quad (4)$$

where G is the medium linear shear modulus, μ is the dynamic viscosity, κ is the ratio of specific heats and R_0 is the radius of the cavitation nucleus. The natural period

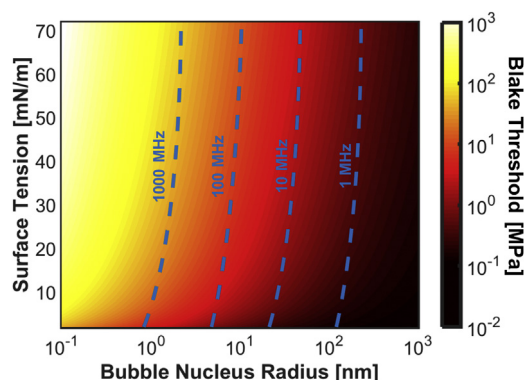


Fig. 5. Blake threshold as a function of bubble nucleus radius and surface tension. The resonant frequencies, calculated in the absence of viscosity and elasticity, of the bubble nuclei are represented by the dashed blue lines.

of oscillation for a nanometer-sized nucleus is approximately $1/2\pi\omega_0 \sim 50$ ps (Fig. 5), much shorter than the tensile period of the histotripsy source. Thus, the tensile period during therapeutic insonation “feels” like a quasi-static change in pressure to the small nuclei.

There are many potential nanoscale nucleus sources. Cosmic rays carry sufficient energy to overcome the bubble nucleation threshold, but are too infrequent to have an impact on the histotripsy cavitation process (244 events/s/m³ in water) (Greenspan and Tschiegg 1967). Harvey et al. (1944) proposed that bubble stabilization can be achieved in the crevices of solid motes. The air/water interface curvature bows into the mote, causing the corresponding Laplace pressure to oppose gas diffusion out of the stabilized nucleus (Apfel 1970; Crum 1979). While observing microscopic impurities in agarose gels, Yount (1979) surmised that a permeable skin coating around the nucleus reduces surface tension and prevents outward diffusion of gas. Nucleus stabilization has been proposed to occur through ionic charges surrounding gas molecules, entities described as charged bubbles or *bubstons* (Bunkin et al. 1996; Sankin et al. 2006). These mechanisms allow indefinite existence of bubble nuclei, but require that the water contain impurities, an almost unavoidable occurrence (Bader et al. 2012b; Flynn 1964). Thus, careful preparation of the water and vessel is required to minimize nuclei that can be expanded by low-amplitude pulses.

There is evidence that bubble nuclei exist *in vivo*. These nuclei can be *extrinsic* or *intrinsic*. Here, we classify *intrinsic* nuclei as the seeds for a bubble that permeate a medium, and would be found under any physiologic conditions. *Extrinsic* nuclei require an inhomogeneity to activate cavitation. Although extrinsic nuclei are found only in discrete locations, their existence is not precluded *in vivo*. The oldest evidence for extrinsic bubble nuclei comes from naval

observations of divers suffering from decompression sickness (Behnke 1945). Bubble activity has been observed in supersaturated *ex vivo* tissue, akin to conditions occurring during the ascension of divers (Crum et al. 2005). Gas-containing organs, such as intestines and lungs, are considered the most susceptible to hemorrhagic injury by ultrasound exposure (Carstensen et al. 2000; Holland et al. 1996). Bubbles have been detected *via* diagnostic ultrasound of kidney stones within the urine (Bailey et al. 2005; Leighton et al. 2008). Cavitation has also been produced in urine and surrounding tissues during shock wave lithotripsy, albeit at significantly higher-pressure amplitudes than the thresholds reported for lungs/intestinal injury (Dalecki et al. 1997; Miller and Gies 2000). The urine contains solid microcrystalline inclusions that may also serve to stabilize nuclei (Fogazzi 1996). Extrinsic nuclei are also activated during tissue heating, such as during thermal HIFU procedures (ter Haar and Daniels 1981; ter Haar et al. 1986). The sources of extrinsic cavitation nuclei triggered at diagnostic ultrasound levels are presumably harbored in crevices such as calcifications within kidney stones, arterial plaque or gas bodies (Kim et al. 2010; Simon et al. 2018). Thus, we see that extrinsic nuclei do not exist without some form of gas-body activation or gas-harboring entity (Miller 2007).

In contrast, intrinsic nuclei are ubiquitous to the medium. Histotripsy-induced cavitation *in vivo* occurs at peak negative pressures between 10 and >25 MPa, similar to pressures at which consistent nucleation is observed in water. Cavitation can be reliably produced at these pressure levels in water-based media, with a sharp decline in the nucleation rate at lower amplitudes. Peak negative pressure thresholds of 26.2 to 29.4 MPa for 1-MHz pulses of 2-cycle duration were observed for water, 5% and 15% gelatin hydrogel tissue phantoms and *ex vivo* canine blood, liver and kidney (Fig. 4). Cavitation was not detected at pressure amplitudes <13.5 MPa for any of these water-based media. For media that are not predominately water based (canine adipose tissue, 1-3 butanediol, and olive oil), the cavitation thresholds do not fall within the range 26.2 to 29.4 MPa (15.4, 35.2 and >36 MPa, respectively).

The tight grouping of the intrinsic threshold for water-based media point to an extremely robust and predictable subset of bubble nuclei that are activated by short-duration pulses, requiring peak negative pressures in excess of 10 MPa to initiate cavitation. Experiments have indicated the ubiquity of this population of bubble nuclei in water, although attempts to isolate and remove these nuclei from water have not been successful (Herbert et al. 2006). Histotripsy exploits this population of nuclei to form cavitation clouds in tissues repeatedly. The close agreement between the intrinsic cavitation threshold of

water-based soft tissues and the intrinsic threshold of water suggests these nuclei are contained within the water component of the tissue. A recent study of histotripsy at the cellular level noted cavitation bubbles were observed to form only in the extracellular space, which has a much larger volume of liquid water compared with the intracellular environment (Vlaisavljevich *et al.* 2016d). However, further work is needed to ascertain the nature of these nuclei and the precise location of nucleation in tissues.

Dependence of intrinsic threshold on histotripsy insonation parameters

The center frequency of single-cycle pulses appears to have minimal influence on the intrinsic cavitation threshold, which increases by <3 MPa from 0.345 to 3 MHz in agar phantoms (Vlaisavljevich *et al.* 2015b). Significant decreases in the cavitation threshold were observed at low-frequency insonations (345–500 kHz) in 90% gas saturated water, likely because of the sources of extrinsic nuclei. The total focal volume, and therefore the total number of insonated nuclei, will be dictated by the frequency and transducer geometry. No significant changes have been observed in the intrinsic cavitation threshold for 500-kHz transducers with f -numbers between 0.5 and 0.9 (Vlaisavljevich *et al.* 2017a).

The histotripsy pulse duration determines the total time the volume is subject to tension, and is also a determining factor in the nucleation of cavitation. *In vitro* and *ex vivo* studies with multiple-cycle histotripsy pulses have indicated that shock-scattering bubble clouds can be generated at lower amplitude compared with single-cycle tensile pulses (10–25 MPa vs. 26–29 MPa) (Maxwell *et al.* 2011b; Vlaisavljevich *et al.* 2014). It is clear that the probability of cavitation increases with longer-duration pulses (Haller *et al.* 2018), although a general quantitative dependency based on the insonation parameters (*e.g.*, fundamental frequency, degree of non-linearity, focal volume) has yet to be established. Li *et al.* (2014b) have measured thresholds for 1-ms pulses in various media. The probability reached near 50% for peak negative pressures of 12.5 MPa in pure water, and even lower in *ex vivo* tissues. Haller *et al.* (2018) observed an increase in the cavitation threshold from 1.1 to 4.6 MPa as the duration of a 1.06-MHz pulse decreased from 1 s to 25 cycles in a 3% agar phantom. More research is needed to reach a general understanding of how these thresholds vary with parameters and measurement systems.

Dependence of intrinsic threshold on medium parameters

Systematic studies have ascertained the dependence of cavitation probability as a function of medium stiffness (Young's modulus) for single-cycle excitations

in vitro (Vlaisavljevich *et al.* 2015b). No correlation was found between the intrinsic threshold and the stiffness of agar phantoms for Young's moduli from 0 to 930 kPa. For *ex vivo* samples, the intrinsic threshold increased 2 MPa for liver (Young's modulus = 8.7 kPa) compared with tendon (Young's modulus = 380 MPa).

In addition to stiffness, other properties of the medium may also play a role in the intrinsic threshold. Cavitation probabilities in excess of 50% have not been measured in more viscous liquids (1,3-butanediol and olive oil, μ = 97 and 84 cP, respectively, vs. 1 cP for water) (Arvengas *et al.* 2011b; Maxwell *et al.* 2013). The temperature dependence of the intrinsic threshold has been explored *in vitro* (Arvengas *et al.* 2011a; Vlaisavljevich *et al.* 2016e). A monotonic decrease in the intrinsic threshold from 29.8 to 14.9 MPa was observed as the temperature of the medium increased from 10°C to 90°C. The measured temperature-dependent cavitation threshold matches the relative changes with temperature predicted by a CNT model (Arvengas *et al.* 2011a; Vlaisavljevich *et al.* 2016e). The surface tension was adjusted to 27.5% of its classical value in the CNT model to match the measured threshold values. Similar reductions in the contributions of surface tension have been suggested for bubble formation during other ultrasound exposures (Herbert *et al.* 2006). Temperature increases also change the surface tension, vapor pressure and viscosity to promote bubble nucleation favorably. It should be noted that simulations of histotripsy-induced bubble activity indicate little change in the maximum bubble diameter over the same range of medium properties corresponding to these temperatures (Bader and Holland 2016).

KNOW WHEN SOMETHING HAPPENS

Apfel's final golden rule, know when something happens, refers to detecting a cavitation event. The type of cavitation event, and therefore the means by which cavitation is detected, is dictated by the cavitation dynamics. One criterion for the onset of inertial cavitation is violent bubble collapse, occurring once the bubble wall velocity approaches the sound speed in the medium (Datta *et al.* 2008; Gruber *et al.* 2014; Haworth *et al.* 2017). Before the inertial collapse, the overwhelming tension in the incident histotripsy pressure waveform forces bubble growth during insonation. Although the expansion can be described to first order in terms of a Blake cavity, the overarching bubble dynamics depend on the form of ultrasound excitation.

Intrinsic-threshold histotripsy

Intrinsic-threshold histotripsy relies on single-cycle pulses with sufficient tension to cause massive expansion

of bubble nuclei in the medium. Single bubbles appear in the focus when the tension is near the intrinsic threshold (Maxwell et al. 2013). As the pressure amplitude is increased, the cavitating volume is increased in proportion to the volume of the focal zone exceeding the intrinsic threshold (Vlaisavljevich et al. 2017a). Once expansion ensues, the bubble dynamics match those of a pure Blake cavity during the acoustic excitation. The bubble grows over the duration of the pulse with sufficient momentum that expansion continues beyond the completion of the excitation (Vlaisavljevich et al. 2015c). Once growth is halted, the bubble will collapse inertially under the ambient pressure.

For the incident waveform to exceed the intrinsic threshold, high-gain sources with f -numbers between 0.5 and 1 are utilized. The focal volume decreases with a reduction in the f -number of the source, thereby refining the region over which bubble activity is generated. The smaller focal volume has the added effect of reducing the waveform non-linearity for a given peak negative pressure (Rosnitskiy et al. 2017). That is, bubbles can be nucleated before the onset of strong non-linearities for intrinsic threshold pulses. The bubble cloud density within the focal regions is also significantly increased for low f -number sources (Vlaisavljevich et al. 2017a).

Although the maximum bubble size is generally proportional to the peak negative pressure, only modest increases in bubble size are observed for tensions in excess of the intrinsic threshold (Fig. 6) (Vlaisavljevich et al. 2015c). The activation of intrinsic nuclei transfers energy from the acoustic wave into the potential and kinetic energy of the bubbles, thus limiting the negative pressure that contributes to the cavitation process (Pishchalnikov et al. 2005). Further increases in amplitude of the acoustic source will result in a greater number of activated nuclei within the focal zone, but bubbles may not experience tension in excess of the intrinsic threshold. A dense bubble cloud is typically nucleated for tensions at the intrinsic threshold (Vlaisavljevich et al. 2015c). The limitation of the maximum bubble size may also be reflective of the strong bubble–bubble interaction (Wang and Brennen 1999) or acoustic

shielding of the incident pulse by bubbles proximal to the transducer (Commander and Prosperetti 1989).

Shock-scattering histotripsy

The f -number of transducers used for shock-scattering histotripsy is generally between 0.7 and 1.0 to compromise between focal gain necessary to activate intrinsic nuclei into extrinsic micrometer-sized scatters and the degree of non-linearity to generate a shock wave (Vlaisavljevich et al. 2017a; Xu et al. 2006a). The resultant waveforms are highly non-linear pulses 3–20 cycles in duration with peak negative pressures between 15 and 25 MPa. Micrometer-sized bubbles are formed within the focal region, though the compressional pressure of the shock wave mitigates the bubble growth rate compared with intrinsic threshold pulses (Maxwell et al. 2011b). In limiting cases where bubble nucleation has a low probability, hundreds or thousands of pulses may be applied before cavitation and the ensuing cloud are achieved. As the bubble grows over the duration of the pulse, a larger portion of the energy in the incident shock wave is backscattered toward the transducer. The bubble–tissue interface acts as a pressure-release boundary condition (Pierce 1989), and the scattered shock wave is inverted with respect to the incident pressure waveform (Fig. 7, left). Once the bubble is sufficiently large, the tension of the total wave field comprising the combined incident and backscattered waves will exceed the intrinsic threshold. The scattered wave nucleates numerous cavitation bubbles between the initial bubble and the histotripsy source. Additional bubbles are nucleated as the subsequent shock scatters from the bubble cloud formed during the previous cycle. The bubble cloud grows against the direction of the incident pulse until the insonation ends or the extent of the bubble cloud reaches the edge of the focal zone (Fig. 7, right). In addition to acoustic sources, scattering bubbles can be alternatively generated *via* pulsed laser sources (Horiba et al. 2017).

Although the compressional component of the pulse can exceed 80 MPa, the bubbles that initiate the shock-scattering process tend to expand throughout the

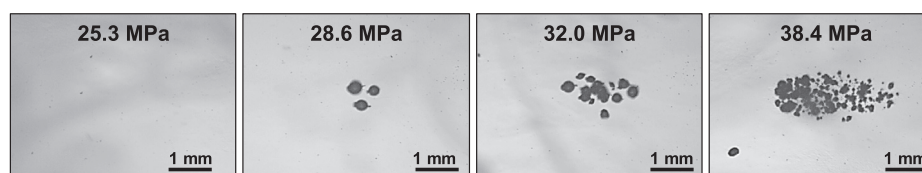


Fig. 6. Bubbles generated by a one cycle histotripsy pulse using a 500 kHz array transducer in an agar phantom as a function of peak negative pressure. Reprinted from *Physics in Medicine and Biology*, vol. 62, Vlaisavljevich et al., Effects of f -number on the histotripsy intrinsic threshold and cavitation bubble cloud behavior, p. 1269. Copyright (2017). doi.org/10.1088/1361-6560/aa54c7. © Institute of Physics and Engineering in Medicine. Reproduced by permission of IOP Publishing. All rights reserved.

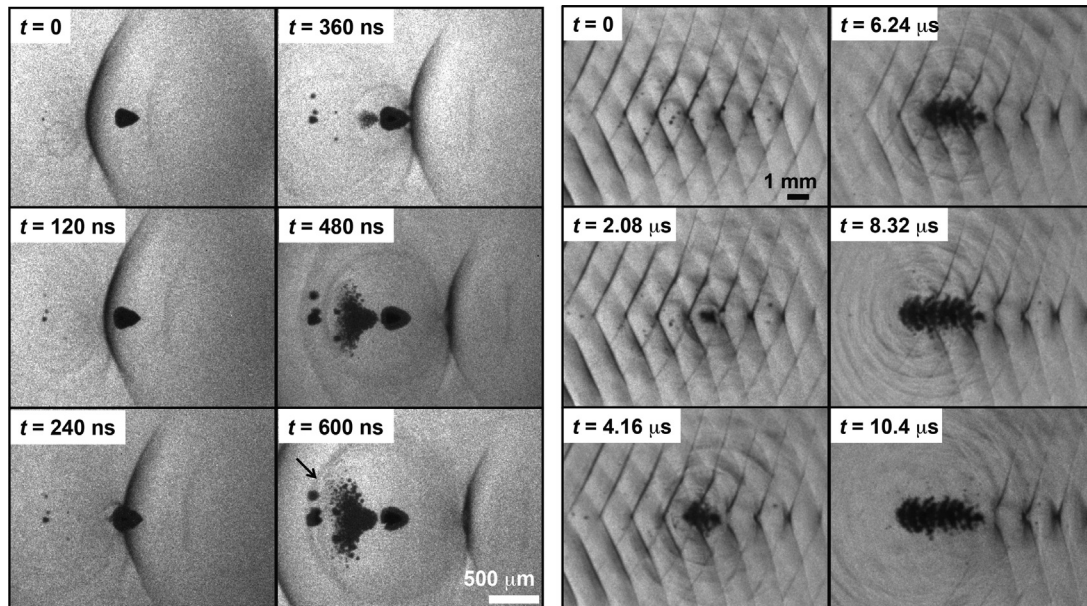


Fig. 7. Left: Initiation of a bubble cloud *via* shock scattering captured by shadowgraphy. Ultrasound propagation is from left to right. The scattering bubble (dark cone, $t = 0$ frame) has been distorted because of the asymmetric incident shock wave (*dark line*). The large size of the scattering bubble compared with the shock thickness and the flattened surface allow strong scattering of the incident shock wave. Furthermore, the pressure release boundary condition of the bubble/gel interface inverts the shock. The scattered, inverted wave nucleates cavitation proximal to the bubble. Right: A shadowgraph sequence revealing formation of a bubble cloud over a 15-cycle pulse. The acoustic propagation is from left to right, but cloud growth forms in the opposite direction during passage of the pulse.

duration of the pulse (Maxwell *et al.* 2011b). The inertia of the large bubble (10–100 μm in diameter) is unable to respond to the short compressional phase of the histotripsy pulse (Ainslie and Leighton 2011). The lack of collapse during the compressional phase of the histotripsy pulse indicates a reversal in the time scales relative to the initial nucleus: The resonant frequency of the nucleus is much larger than the fundamental frequency of the incident histotripsy pulses (see Fig. 5). During expansion, the effective resonant frequency of the bubble decreases significantly compared with the high frequencies associated with the shock wave.

The strength of shock scattering is strongly dependent on the shape of the initial scatterer. The shocked, asymmetric excitation may deform the bubble from its spherical shape into a hemisphere or a concave form, as depicted in Figure 7 (Maxwell *et al.* 2011b). Distortion of the bubble shape can occur because of the pressure differential across the bubble when the shock is incident on one side, and is related to jetting phenomena (Brujan *et al.* 2001). The distortion is consistent with numerical models of shock wave-induced bubble behavior (Johnsen and Colonius 2008). Scattered fields from hemispherical bubbles are cone shaped, and have the largest field strengths within $\lambda/2$ from the scattering interface (λ is the acoustic wavelength of the histotripsy

pulse fundamental frequency) (Maxwell *et al.* 2010a). Consequently, new bubbles are nucleated in close proximity to the initial bubble. The peak negative pressure of a scattered field from a spherical bubble is significantly reduced compared with that from a hemispherical bubble (Maxwell *et al.* 2010a). Bubble clouds can still form, however, when the pulse scattered from a spherical bubble interferes constructively with the subsequent tensile phase of the shock-scattering histotripsy pulse.

The formation of each layer of bubbles serves as a new interface from which the shock wave scatters, causing the bubble cloud to grow. The axial growth rate is approximately $\lambda/3$ per acoustic cycle once shock scattering is initiated (Maxwell *et al.* 2011b). The closest pressure peak of constructive interference would occur at a distance $\lambda/4$ from the bubble surface for a linear waveform, and increase to $\lambda/2$ for a sawtooth waveform. Depending on the degree of waveform distortion and the region over which new bubbles occur during a cycle, the peak tension can occur anywhere between these values. The non-linear field decreases quickly with distance beyond the focal zone (Rosnitskiy *et al.* 2017), and is the limiting factor for the axial extent of the bubble cloud. The lateral growth of the cloud (*i.e.*, perpendicular to the central axis of the histotripsy source) is largely independent of the pulse duration and is limited in extent by the beam width.

Boiling histotripsy

Thermal ablation with focused ultrasound denatures tissue over the course of seconds or minutes. The heating rate Q is proportional to the intensity of the sound field and attenuation in the focus (Hallaj and Cleveland 1999). Shock waves, consisting of a series of harmonics, dissipate energy much more rapidly over a distance compared with a linear waveform (Hamilton and Blackstock 1998). Under the condition that heating occurs quickly enough, the tissue can boil and cause mechanical disintegration before producing any noticeable thermal effects. Such is the case with boiling histotripsy.

Shock-induced heating during a boiling histotripsy pulse will alter the properties of the medium and increase the likelihood of bubble generation. The time for onset of boiling (t_b) can be estimated under the assumption that the heating rate is sufficiently large to neglect thermal diffusion (Canney et al. 2010):

$$t_b = \frac{\Delta T C_v}{Q} \quad (5)$$

Here C_v is the heat capacity, and ΔT is the temperature difference required to reach a boiling temperature, 100°C. When a shock wave is fully formed, the heating rate can be estimated via weak shock theory as $Q_S = \beta f_0 A_S^3 / 6c_0^4 \rho^2$ (Canney et al. 2010), where A_S is the shock wave amplitude, and f_0 is the fundamental frequency of the insonation. A shocked waveform increases the heating rate significantly compared with a linear pressure waveform of the same pressure amplitude, allowing boiling to occur within milliseconds. However, the onset of boiling requires bubble nucleation. Increases in temperature decrease the threshold tension required to achieve intrinsic nucleation, which decreases to ~15 MPa at 90°C (Vlaisavljevich et al. 2016e). Rapid heating may result in local tissue temperatures beyond 100°C, further reducing the threshold

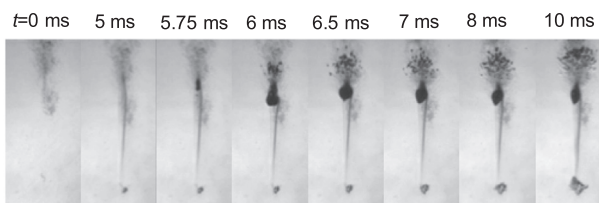


Fig. 9. Temporal sequence of millisecond boiling at the focus of a 1-MHz boiling histotripsy transducer. As heat is deposited, a shadow appears as a result of changes in the index of refraction around the focus ($t=5$ ms). At $t=5.75$ ms, a boiling bubble occurs, and a cavitation cloud appears behind the bubble over the next 0.5 ms as a result of scattering. Ultrasound propagation is from top to bottom.

until a nucleus is generated. The stochastic processes associated with the activation of intrinsic nuclei in heated media may explain variability of the time-to-boil observed in experiments (Canney et al. 2010).

The parameter space over which boiling can be initiated within 10 ms (a reasonable time to avoid thermal effects to the tissue) is well defined in soft tissues, as depicted in Figure 8a. With respect to the insonation parameters, the calculated time to boil is dependent on the fundamental frequency and shock amplitude. Given the complex relationship between the intrinsic nucleation and temperature (Vlaisavljevich et al. 2016e), a correlation between the time-to-boil and the peak negative pressure may also exist.

Similar to shock-scattering histotripsy, the large boiling bubble within the focal zone can act to scatter the incident shock wave. Indeed, a structured bubble cloud akin to that documented in shock-scattering histotripsy has been observed to form proximal to the boiling bubble (Fig. 9) (Khokhlova et al. 2011; Pakh et al. 2017). The layers of the bubble cloud were separated by $\lambda/2$. Numerical calculations of the interactions of shock waves with bubbles 100–500 μm in diameter confirm that regions of constructive interference occur at

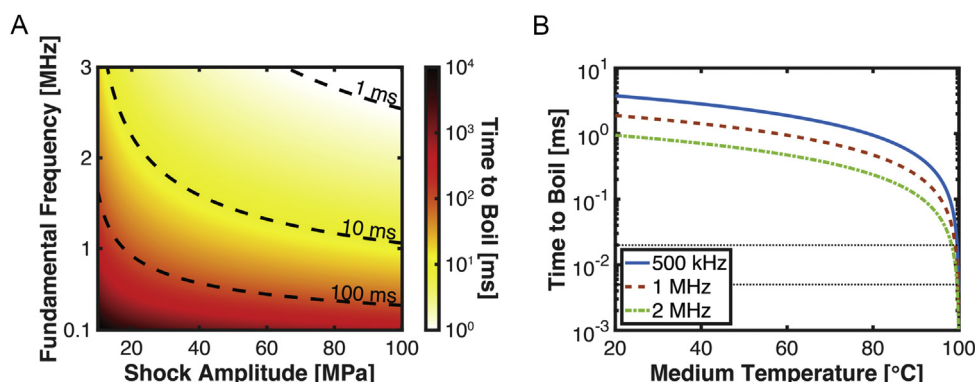


Fig. 8. A: Time-to-boil for shock-induced heating in soft tissues. B: Time-to-boil as a function of the temperature of the medium. The center frequency of the insonation is noted in the legend, and a 100-MPa shock amplitude was assumed. The dashed black lines span 5–20 μs in the right panel represent the typical duration of a shock-scattering histotripsy pulse (Khokhlova et al. 2015; Maxwell et al. 2012). The specific heat per unit mass was $3.5 \times 10^6 \text{ J/m}^3 \cdot \text{C}$.

a distance of $\lambda/2$ from the bubble (Pahk *et al.* 2017). The discrete layering of the bubble cloud in boiling histotripsy differs from the bubble cloud layering observed in shock-scattering histotripsy because of variations in the amplitude and degree of non-linearity between the insonation schemes.

Although shock-scattering cavitation can occur during boiling histotripsy, boiling is unlikely during intrinsic-threshold or shock-scattering histotripsy. For a 1-MHz fundamental frequency insonation, shock amplitudes in the range 7 to 16 GPa would be required to initiate boiling during a shock-scattering histotripsy pulse based on eqn (5). Even at an elevated temperature (*e.g.*, thermal HIFU used in combination with histotripsy), boiling during shock-scattering histotripsy pulse is unlikely when the initial medium temperature is less than 99°C (Fig. 8b).

Bubble dynamics modeling

To predict the histotripsy-induced cavitation dynamics, and therefore the potential for tissue liquefaction, a Rayleigh–Plesset-like equation can be solved with a histotripsy excitation forcing function $p_a(t)$:

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \left[\left(p_0 + \frac{2\sigma}{R_0} - p_v \right) \left(\frac{R_0}{R} \right)^{3\kappa} + p_v - \frac{2\sigma}{R} - \frac{4\eta\dot{R}}{R} - p_0 - p_a(t) \right] / \rho \quad (6)$$

The diacritic dot denotes the temporal derivative. Detailed derivation of the Rayleigh–Plesset equation, as well as derivation of modified models that incorporate fluid compressibility, is provided elsewhere (Flynn 1964; Leighton 1994; Prosperetti and Lezzi 1986). For histotripsy-induced cavitation, the Rayleigh–Plesset model should be extended to incorporate the effects of elasticity to mimic a viscoelastic tissue. An early version of bubble oscillations in viscoelastic medium were modeled with linearized Kelvin–Voight elasticity (Yang and Church 2005). Calculations based on the Kelvin–Voight elastic model have been reported to be in qualitative agreement with experimental observations of shock scattering-induced bubble behavior (Bader and Holland 2016). Further refinement of the elastic model through incorporation of a finite-strain neo-Hookean elasticity (Vlaisavljevich *et al.* 2015c, 2016d) or the strain-hardening Fung elastic model (Movahed *et al.* 2016) have also been reported to be in qualitative agreement with experimental observations.

In general, the highly non-linear nature of the Rayleigh–Plesset equation necessitates numerical solutions, and bubble expansion by histotripsy pulses has been explored numerically by several groups (Bader and Holland 2016; Mancia *et al.* 2017; Maxwell *et al.* 2013; Movahed *et al.* 2016; Vlaisavljevich *et al.* 2015c;

Pahk *et al.* 2015). The tensions well in excess of the Blake threshold for histotripsy pulses simplify the Rayleigh–Plesset equation significantly, and the maximum bubble size can be predicted with high accuracy for a single-cycle excitation analytically (Bader and Holland 2016):

$$R_{\text{MAX}} = \left[R_0 + \sqrt{\frac{2P_0 p_0 \xi}{9\rho}} \tau \right] \left[\frac{\xi p_0 P_0}{3p_{\text{EFF}}} + 1 \right]^{1/3} \quad (7)$$

Here ξ and τ are defined in Holland and Apfel (1989). The effects of surface tension, viscosity and inertia are accounted for in eqn (7). The analytic model has also been extended to account for finite-strain and strain-hardening media (Bader 2018).

In addition to its simplicity, the analytic model provides a means to identify appropriate contributions of the insonation scheme and medium properties to the overall bubble dynamics. The parameter τ denotes duration for forced bubble expansion, and can be expressed as $1/f$ when the peak negative pressure is much greater than the Blake threshold (Leighton 1994). The limiting value for τ indicates the maximum bubble size is inversely proportional to the fundamental frequency of the insonation, consistent with experimental observation (Vlaisavljevich *et al.* 2015c). The time-averaged pressure experienced by the bubble during its growth phase is expressed as $p_0 \xi/3$. The term p_{EFF} in eqn (7) represents the effective overpressure that arrests bubble growth after tension is no longer applied. The form of the effective pressure depends on the insonation type, as the contribution of compressional components varies between histotripsy insonation schemes. For a shock-scattering histotripsy pulse, the maximum bubble size depends roughly on the square root of the peak negative pressure due to the form of the effective pressure. In contrast, the bubble size increases nearly linearly with the peak negative pressure for intrinsic threshold excitations. These results indicate that nuclei grow larger by intrinsic threshold insonations compared with a shock-scattering histotripsy pulse.

Care should be taken when applying eqn (7) to predict bubble growth. The pressure experienced by the bubble is assumed based on a pre-defined peak negative pressure, whereas the maximum bubble diameter saturates when the applied tension exceeds the intrinsic medium threshold, as discussed in the section Intrinsic-Threshold Histotripsy in this manuscript (Vlaisavljevich *et al.* 2015c). For shock-scattering histotripsy, eqn (7) predicts the growth of a bubble only during the first cycle of the pulse. Beyond the first cycle, the growth rate of the bubble will decrease and depends on the time-averaged acoustic pressure. These analytic computations

can be utilized to estimate the onset of shock scattering through classical scattering theory (Anderson 1950) and the axial extent of the cavitation based on the known shock scattering-induced bubble cloud growth rate (Bader and Holland 2016).

Properties of the medium that influence the bubble dynamics

For histotripsy excitations, the maximum bubble size is relatively independent of the medium surface tension (Bader and Holland 2016; Maxwell et al. 2013). The medium viscosity and elasticity contribute to the degree of bubble expansion. Indeed, a significant reduction in the maximum bubble diameter has been observed in *in vitro* agar phantoms with increasing Young's modulus over the range 0 (water) to 570 kPa (Vlaisavljevich et al. 2015c). For a given elasticity, a lower-frequency excitation resulted in larger bubble expansion because of the extended duration of the tensile phase, as depicted in Figure 10. However, the threshold for bubble nucleation with single-cycle pulses was nearly independent of the tissue stiffness and the insonation frequency (Vlaisavljevich et al. 2015b).

Although medium stiffness has minimal contribution to the intrinsic threshold, it does have ramifications

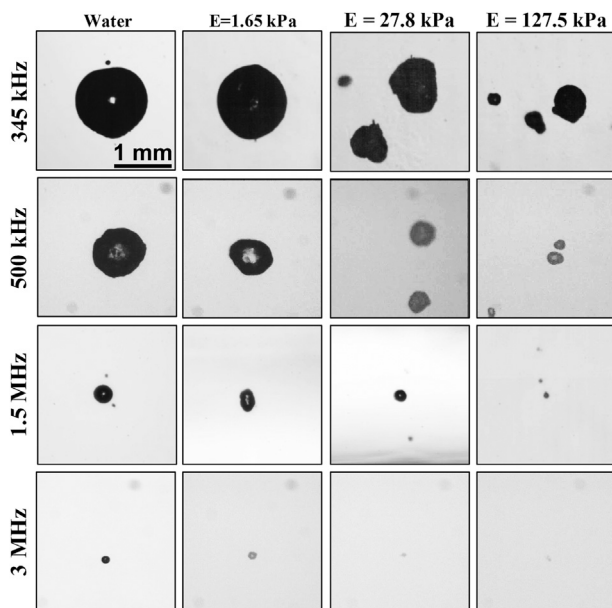


Fig. 10. Optical images of bubbles produced by histotripsy pulses inside agarose tissue phantoms of increasing Young's modulus. Images reveal a decrease in bubble size with increasing frequency and increasing medium stiffness. Reprinted from *Physics in Medicine and Biology*, vol. 60, Vlaisavljevich et al., Effects of tissue stiffness, ultrasound frequency and pressure on histotripsy-induced cavitation bubble behavior, p. 2271. doi.org/10.1088/0031-9155/60/6/2271. Copyright (2015). © Institute of Physics and Engineering in Medicine. Reproduced by permission of IOP Publishing. All rights reserved.

for the microscale bubble dynamics. This is reflected in the pressure at the bubble wall due to medium properties during the bubble oscillation. Surface tension and viscous pressure terms are inversely proportional to the bubble size, and decrease as the bubble undergoes massive expansion (Fig. 11). The elastic pressure contribution for a finite-strain elastic model, however, approaches a maximum value when the bubble size is much larger than that of the nucleus (Gent 1996). The elastic pressure term therefore acts as an additional overpressure to suppress bubble growth, thereby reducing the maximum bubble diameter.

Sustained bubble dynamics

The discussion thus far has focused on bubble behavior over the course of a single ultrasound pulse. Tissue ablation requires the application of tens to hundreds of pulses for complete liquefaction of the focal zone (Wang et al. 2012). For all forms of histotripsy, bubbles appear to persist well after the application of the histotripsy pulse. Remnants of bubble clouds have been observed optically up to 30 ms after the therapeutic excitation (Prieur et al. 2015; Xu et al. 2007) and up to 50 ms post-insonation acoustically (Bader et al. 2018). A potential cause of these persistent bubble clouds is gas diffusion during bubble expansion (Bader and Bollen 2018). The infiltration of gas dissolved in the medium into the bubble will increase its equilibrium size, thereby slowing the passive dissolution process (Church 1989). Fragmentation of the bubble during an inertial collapse can also proliferate daughter bubbles (Leighton 1995). The remnant nuclei produce a “memory effect,” whereby bubbles at the same physical location are repeatedly excited and a similar structure of the cloud appears from pulse to pulse.

The peak negative pressure necessary to produce inertial cavitation with a subsequent pulse is lowered in the presence of a remnant bubble cloud (Xu et al. 2007). The reduced threshold for persistent bubble clouds has been utilized for strategies of tissue erosion by applying a higher-amplitude “bubble cloud-initiating” pulse followed by lower-amplitude “maintenance” pulses (Xu et al. 2006b). The expansion of bubbles within the cloud shields the acoustic pulse from the focal zone, with the nuclei closest to the transducer undergoing greater expansion than those in the distal part of the focus. The asymmetric axial expansion produces a comet-shaped head and tail of the bubble cloud.

With each pulse, bubble activity contributes to the eventual mechanical breakdown of the tissue. Cellular stress/strain *via* bubble oscillations appears to be a primary mechanism of histotripsy-induced ablation (Vlaisavljevich et al. 2016d). Other mechanisms, such as acoustic atomization (Simon et al. 2012), may also

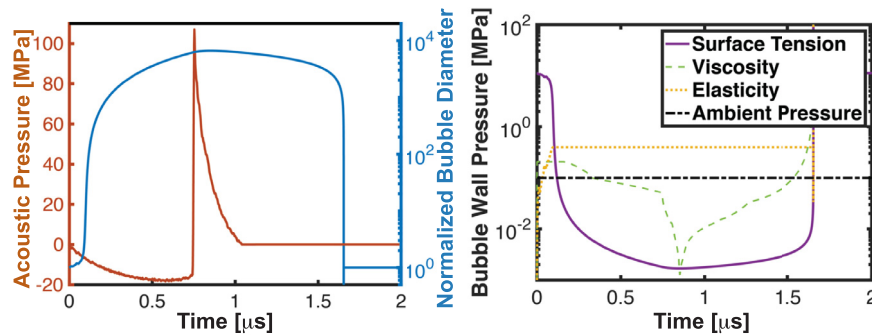


Fig. 11. Left: Calculated response of a 20-nm-diameter bubble (right-hand axis) to a single cycle of a shock-scattering histotripsy pulse (left-hand axis) via the Yang/Church model (Yang and Church 2005). Right: External pressure forces acting on bubble wall during the excitation due to surface tension, viscosity, elasticity, and the ambient pressure (0.1 MPa). The values of the medium properties were as follows: surface tension, $\sigma = 56$ mN/m, dynamic viscosity, $\mu = 0.005$ kg/m-s, and elastic modulus, $E = 100$ kPa. The exact pressure at the bubble wall will change for variations in the cavitation process (e.g., bubble size, medium properties, histotripsy excitation), though the trends will remain (Bader 2018).

contribute to tissue breakdown. The strain necessary to rupture a cell is a function of time of the applied strain (Hategan *et al.* 2003; Li *et al.* 2013). Large cellular deformations (more than 150% stretching) are generated in the presence of histotripsy bubbles (Vlaisavljevich *et al.* 2016d). Although cells in the proximity of the bubble appear to be lysed within the first pulse, cell bisection or complete removal at the edge of the focal zone occurs over the course of several pulses. Thus, sustained activity throughout the focal zone is the hallmark for complete liquefaction of the target tissue (Vlaisavljevich *et al.* 2016d).

Insonation of residual bubbles causes strain at discrete locations within the focal zone during a pulse, but incomplete liquefaction of the target tissue (Wang *et al.* 2012). Lesions may progress toward complete disintegration through several potential means. New nuclei within the cloud may be excited, causing damage in new locations (Maxwell *et al.* 2010b). Bubbles in the same location may damage their surroundings, and their dynamics change from pulse to pulse to grow and damage a greater area (Movahed *et al.* 2017). A third possibility is that the remnant nuclei undergo translational motion between pulses, increasing the spatial extent of damage.

Changes in the sustained bubble dynamics occur in boiling histotripsy as well. The onset of boiling is dependent on the number of applied pulses. Boiling coincides with the predicted time-to-boil for a single pulse (Canney *et al.* 2010), but occurs before the predicted time in subsequent pulses (Khokhlova *et al.* 2011). The reduction in the time-to-boil can be attributed to heat deposition between pulses, despite the relatively low pulse repetition frequency (1 Hz, 0.01 duty factor). Over the course of the therapeutic application, the extent of the boiling bubbles encompasses the focal volume.

AMENDMENTS TO APFEL'S GOLDEN RULES

In this section, ongoing and proposed studies are discussed for promoting tissue-stressing bubble activity through the manipulation of the acoustic parameters and nucleation environment and monitoring the histotripsy process. These processes are proposed as a modification of Apfel's golden rules to accentuate histotripsy-induced cavitation activity: Change thy sound field, change thy liquid and quantify when everything happens.

Change thy sound field

One approach to improve histotripsy therapy for various clinical applications is to modulate the applied acoustic field to control histotripsy-induced bubble dynamics. Boiling and shock-scattering histotripsy rely on multiple processes to induce the bubble activity necessary for tissue fractionation. In contrast, nucleation through tensile-only pulses is a single process. The bubble cloud dimensions (size, shape, location, bubble density) can be predictably changed by altering the geometry of the therapy transducer (Vlaisavljevich *et al.* 2017a). Furthermore, a stochastic model can be used to predict the lesion dimensions for a given ultrasound field and number of pulses applied to a region (Lin *et al.* 2014b; Maxwell *et al.* 2013). Such modeling provides a theoretical basis for design of devices and therapy planning in the intrinsic threshold regime.

Dual-frequency transducers for intrinsic nucleation allow the targeting of deep tissue targets (Lin *et al.* 2015) or incitation of specific bubble behaviors (Vlaisavljevich *et al.* 2015c). By use of a multifrequency source, nearly monopolar waveforms with a single compression or tension phase can be generated in the focus (Lin *et al.* 2014a). No appreciable mechanical effects are

produced for 100-MPa monopolar compressive pulses, whereas tensile monopolar pulses with a peak negative pressure of 30 MPa consistently nucleated bubble clouds in tissue-mimicking phantoms. Monopolar pulses may have advantages for histotripsy therapy applications, such as more precise bubble cloud generation because of decreased nucleation through scattering, and a reduced thermal profile. Additionally, monopolar excitations can be utilized as a tool to investigate the underlying histotripsy nucleation processes (Vlaisavljevich et al. 2016b).

Although intrinsic-threshold histotripsy represents the most predictable form of histotripsy therapy, it is often not feasible to generate the tension necessary for intrinsic nucleation. Multiple-cycle acoustic pulses can be used to generate shock-scattering or boiling histotripsy bubble clouds at tensions less than the intrinsic threshold. The dependence of the shock-scattering bubble cloud formation on the insonation parameters (e.g., pulse repetition frequency, degree of non-linearity, pulse duration) (Maxwell et al. 2011b; Vlaisavljevich et al. 2014) indicates that pulsing strategies can be optimized for tissue ablation. Yoshizawa et al. (2012) devised a two-step amplifier that enhances shock scattering by controlling the degree of non-linear waveform distortion. Another insonation scheme from the same group employed negative-enhanced or positive-enhanced pulses to promote sparse bubble growth or shock scattering, respectively. Bubble clouds were readily nucleated with insonation schemes of negative-enhanced pulses followed by positive-enhanced pulses. When the order was reversed, bubble cloud formation was suppressed because of a lack of initial scatterers from the negative-enhanced wave. These shock-induced bubble clouds were used to enhance thermal HIFU heating.

Frequency is another parameter that should be considered for efficacious tissue fractionation. The intrinsic threshold is not strongly dependent on the driving frequency (Vlaisavljevich et al. 2015b). However, the maximum bubble size, and therefore extent of strain-induced damage (Bader 2018), is inversely proportional to frequency (Fig. 10). Other variables, such as the bubble dynamics, rate and effectiveness of ablation, potential for collateral damage, target depth and pre-focal cavitation are all affected by the choice of frequency. The choice of histotripsy insonation frequency should therefore weigh the likelihood of inducing the intended bubble activity with these potential undesirable outcomes.

Sustaining the mechanical activity necessary for histotripsy-induced ablation requires the activation of intrinsic nuclei for each histotripsy pulse. The primary limitation in activating new intrinsic nuclei is bubble clouds that persist between pulses (Xu et al. 2007). Insonation of these extrinsic nuclei causes strain at discrete locations within the focal zone and incomplete

liquefaction of the target tissue (Wang et al. 2012). Pulse repetition frequencies of 1 to 10 Hz are required to allow complete dissolution of bubbles between excitations (Maxwell et al. 2013; Wang et al. 2012). Histotripsy treatment planning thus requires a compromise between efficacy and duration. Low-amplitude “bubble-deleting” pulses interleaved with histotripsy pulses have shown promise in the removal of residual bubbles in both *in vitro* phantom studies (Duryea et al. 2015) and model urinary stones (Roberts et al. 2015). Bubble-deleting pulses may act to force bubble coalescence through secondary Bjerknes forces (Duryea et al. 2014), with the large bubbles rising under buoyancy out of the focal zone. Alternatively, the bubble-deleting pulses may force the residual bubbles into different parts of the tissue *via* primary Bjerknes forces, allowing uniform bubble activity throughout the focal zone (Acconcia et al. 2013).

Change thy liquid

Another approach to enhancing histotripsy is the modulation of nuclei within the target tissue through the introduction of exogenous agents (Fig. 12). Ultrasound

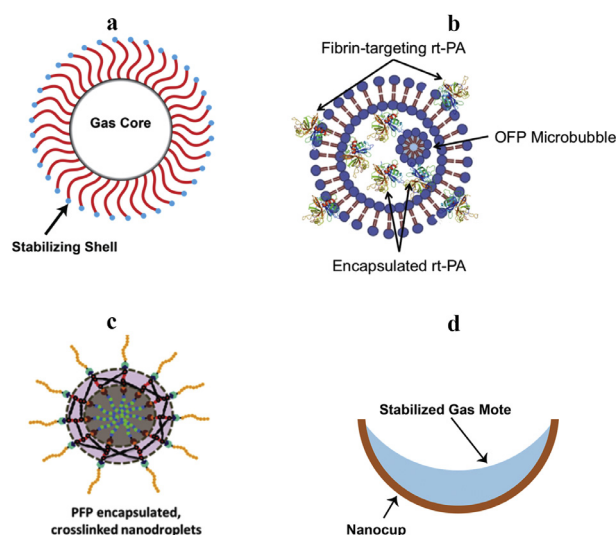


Fig. 12. Overview of exogenous nuclei utilized in histotripsy. (a) Microbubble contrast agent, typically composed of a high-molecular-weight gas (e.g., C_3F_8) surrounded by a stabilizing shell, such as a lipid. (b) Schematic of echogenic liposome loaded with the thrombolytic drug recombinant tissue plasminogen activator and octafluoropropane (OFP) gas microbubbles. A portion of the thrombolytic is encapsulated within the liposome. The remaining portion is intercalated within the lipid bilayer, exposing the finger domain to target fibrin. On exposure to a histotripsy pulse, the encapsulated bubble will expand, locally releasing the thrombolytic. (c) Polymer-encapsulated nanodroplets containing perfluoropentane (PFP) quid core used to lower the cavitation nucleation threshold. (d) Nanocup used to entrap and stabilize gas to act as an extrinsic cavitation nucleus.

contrast microbubbles were utilized in the earliest formulations of histotripsy to enhance cavitation activity (Tran *et al.* 2003). Although microbubbles can enhance the efficacy of ablation with focused ultrasound sources (McDannold *et al.* 2006, 2013, 2016) and display histotripsy-induced dynamics similar to those of intrinsic nuclei (Bader *et al.* 2016a), bubble nucleation would be restricted to vasculature (Kwan *et al.* 2015). Exogenous nuclei reduce the threshold for histotripsy-induced bubble nucleation (Vlaisavljevich *et al.* 2013a), thereby reducing the overall bubble size (Bader 2018). The change in bubble dynamics with exogenous nuclei consequently reduces the efficacy of tissue ablation (Vlaisavljevich *et al.* 2016a). Microbubbles are no longer necessary to generate cavitation because of the advances in histotripsy transducer and amplifier technology (Hall and Cain 2006). However, there remain some potential advantages of using exogenous agents for targeted ablation.

Nanodroplets have been explored for several years as a means of inducing acoustically driven cavitation (Kripfgans *et al.* 2000). Nanodroplets are composed of a metastable perfluorocarbon liquid surrounded by a stabilizing shell. A high-amplitude acoustic pressure transitions

the liquid to a gas, producing a histotripsy-friendly bubble (Fig. 13) (Apfel and Roy 1983; Radhakrishnan *et al.* 2016). Formulations of nanodroplets specifically to target micro-metastases have been developed (Yuksel Durmaz *et al.* 2014) and tested *in vitro*. Initial studies found a significant decrease in the threshold for bubble cloud formation with ~200-nm-diameter nanodroplets (~10 MPa) compared with no droplets (~28 MPa) for 500-kHz insonations (Vlaisavljevich *et al.* 2013a). The microtriopsy-mediated droplet transition threshold increases with frequency (Vlaisavljevich *et al.* 2015a), in contrast to measurements with non-histotripsy insonations (Kripfgans *et al.* 2000). For nanodroplet-mediated histotripsy (NMH), homogenous nucleation of the perfluorocarbon liquid occurs *via* homogenous tensile nucleation (Vlaisavljevich *et al.* 2015a), whereas heating *via* superharmonic focusing is a mechanism for non-histotripsy droplet transition (Li *et al.* 2014a; Shpak *et al.* 2013). Liquefaction of agar phantoms (Vlaisavljevich *et al.* 2013a) and cancer tumor spheroids (Aydin *et al.* 2016) can be achieved with NMH. A larger number of pulses are required for complete liquefaction of the focal zone for NMH than for insonation without droplets, indicating the smaller size of bubbles nucleated from droplets compared with intrinsic nuclei. Sustainable droplet nucleation is dependent on the boiling temperature of the perfluorocarbon (Vlaisavljevich *et al.* 2016a). At peak negative pressures above the threshold for bubble cloud formation, cavitation is sustained for only 10 pulses for droplets formed with perfluoropentane (boiling point of 29°C). Robust bubble clouds were observed for droplets formed with perfluorohexane (boiling point of 56°C) up to 1000 pulses, likely because of the re-condensation of the gas into a liquid. Thus, higher-boiling-point perfluorocarbon droplets have an advantage for sustained NMH ablation.

Echogenic liposomes (ELIPs) are theragnostic agents that can be functionalized for targeted ultrasound contrast (Tiukinhoy-Laing *et al.* 2007) and localized drug delivery (Shaw *et al.* 2009; Smith *et al.* 2010). Gas microbubbles entrained within ELIPs can act as cavitation nuclei (Bader *et al.* 2015b), and may serve as sites for shock scattering to initiate bubble cloud formation (Tran *et al.* 2003). Thrombolytic-loaded ELIPs with octafluoropropane gas microbubbles (OFP t-ELIPs) were utilized to test the efficacy of clot lysis in combination with histotripsy (Bader *et al.* 2016a). An improvement in thrombolytic efficacy was observed for the combination of histotripsy and OFP t-ELIPs compared with histotripsy alone. The degree of cavitation activity was similar when comparing treatment arms with and without OFP t-ELIPs.

An alternative scheme for exogenous nucleation is the stabilization of gas within moles on hydrophobic particles (Apfel, 1970 Crum 1979). Through an interfacial seed polymerization method (Okubo *et al.* 2001),

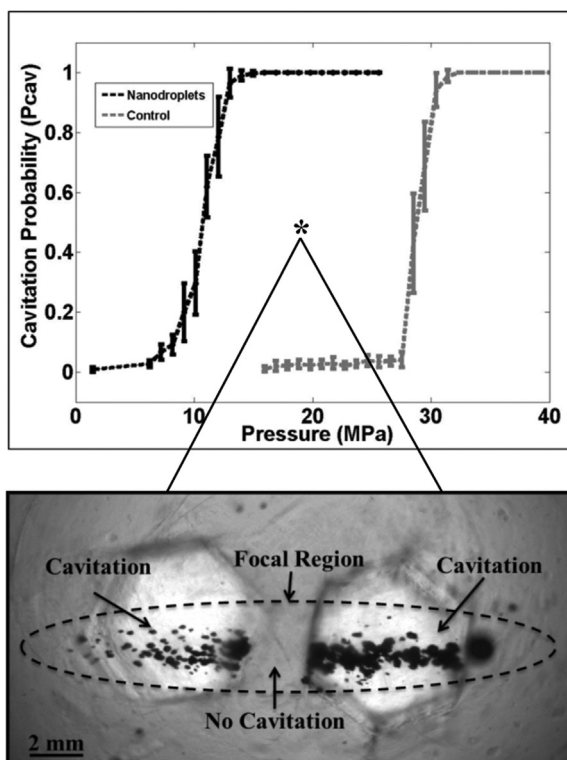


Fig. 13. Nanodroplet-mediated histotripsy. (a) Perfluorocarbon nanodroplets significantly reduce the histotripsy intrinsic threshold, allowing for (b) selective generation of cavitation only in regions containing the targeted nanodroplets. Reprinted from *Theranostics*, vol. 3, Vlaisavljevich *et al.*, Nanodroplet-mediated histotripsy for image-guided targeted ultrasound cell ablation, p. 851. doi:10.7150/thno.6717 (2013).

nanocups have been manufactured from polystyrene spheres (Kwan et al. 2015). The hydrophobic surface of the nanocup harbors a gaseous core that will undergo inertial cavitation at peak negative pressures between 0.5 and 1 MPa (Kwan et al. 2015). Because of their relatively small size compared with microbubbles (100–500 nm vs. 0.6–10 μm) (Graham et al. 2014), nanocups and droplets can efficiently penetrate leaky tumor vasculature (Kwan et al. 2015). Other potential applications for nanocups include transdermal transport of vaccines (Bhatnagar et al. 2016) and delivery of oncolytic vaccinia viruses (Myers et al. 2016).

In addition to adding exogenous nuclei, the nucleability of a medium can be altered through temperature. Heating reduces tissue stiffness through the denaturing of collagen and other proteins, particularly in tissue resistant to mechanical activation (Vlaisavljevich et al. 2015d). Stiff tissues naturally resistant to histotripsy can be softened *via* heating to enhance their susceptibility to histotripsy-induced mechanical damage. Thermal pre-treatment can also be used for reducing the nucleation pressure required for generating microtripsy bubble clouds (Vlaisavljevich et al. 2016e), with these two approaches likely complementing each other. Similar changes in the tissue occur during boiling histotripsy, as

a small section of the tissue near the center of the focus is heated to at least 100°C.

Quantify when everything happens

The final means for improving histotripsy therapy is the advancement of image guidance methods. B-Mode echogenicity is the most ubiquitous parameter for histotripsy image guidance and assessment of treatment efficacy (Hall et al. 2007a). The bubble cloud appears hyper-echoic on a B-mode ultrasound image, whereas liquefied tissue appears hypo-echoic (Fig. 1) (Hall et al. 2005, 2007a). Significant variability between patients has been noted, preventing quantification of B-mode images (Hall et al. 2007a). This has prompted the testing of alternative imaging modalities for histotripsy. These imaging methods can be categorized as real-time assessment of bubble activity for image guidance (Fig. 14) or assessment of the treatment efficacy through bubble-mediated changes in tissue structure.

Quantify bubble activity

There is some advantage to quantifying bubble cloud activity during insonation, as the morphology of the liquefaction zone changes with time (Vlaisavljevich et al. 2016c), and the immediate physiologic responses

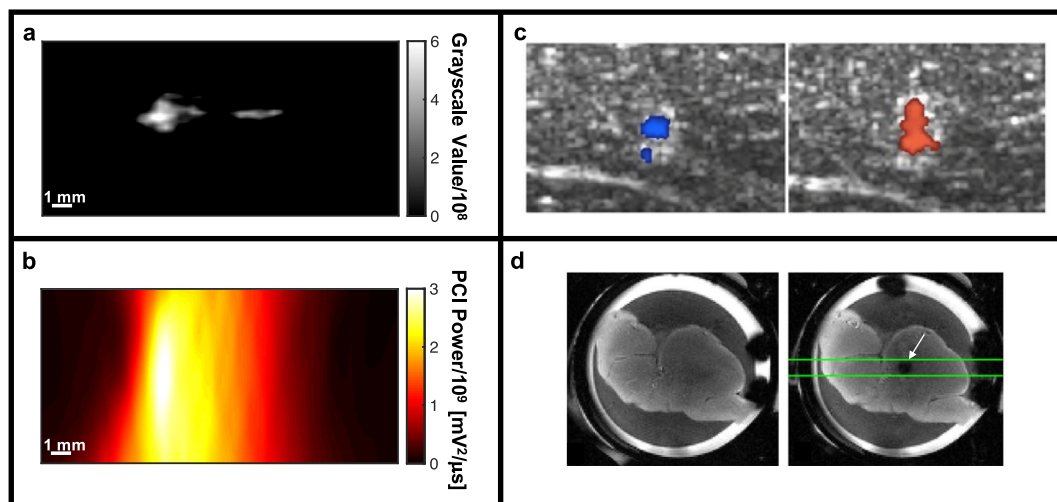


Fig. 14. Summary of imaging modalities for real-time assessment for histotripsy image guidance. (a) B-Mode imaging of hyper-echoic bubble cloud *via* changes in gray-scale value (histotripsy pulse propagating from left to right in the image). (b) Passive cavitation imaging (PCI) maps acoustic emissions generated by the bubble cloud spatially (histotripsy pulse propagating from left to right in the image). (c) Color Doppler images acquired during histotripsy liquefaction of *ex vivo* porcine liver, indicating movement both toward (left image) and away from (right image) as a result of coherent motion associated with translation of the bubble cloud (Miller et al. 2016). (d) *Ex vivo* porcine liver sample before histotripsy excitation imaged with a spin-echo imaging sequence (left frame), and just after histotripsy excitation with a cavitation-sensitive 2-D EPI sequence (white arrow, right frame) (Allen et al. 2015). Panel C reprinted with permission from *IEEE Transactions on Ultrasonics, Ferroelectrics and Frequency Control*, vol. 63, Miller et al., Bubble-induced color Doppler feedback for histotripsy tissue fractionation, p. 408. doi:10.1109/TUFFC.2016.2525859. Copyright (2016). © IEEE. Panel D reprinted with permission from *Magnetic Resonance in Medicine*, vol. 76, Allen et al., MR-based detection of individual bubble clouds form in tissue and phantoms, p. 1486. doi: 10.1002/mrm.26062. Copyright (2015). © John Wiley and Sons, Inc.

may mask imaging assessment of ablation zone extent (Zhou 2011). For thermal ablation, magnetic resonance (MR) thermometry can be utilized to generate maps of thermal dose to estimate the therapeutic efficacy *in situ* (Rieke and Butts Pauly 2008). Histotripsy pulses cause minimal tissue heating (Kieran *et al.* 2007), and an equivalent means for monitoring the “mechanical dose” of the therapy is required. During the oscillation of a bubble, a portion of the mechanical work is transformed into acoustic emissions (Leighton 1994). Acoustic emissions can serve as a surrogate for the mechanical action of cavitation (Bader *et al.* 2012a), and to quantify the type of bubble activity. Stable cavitation can be characterized by acoustic emissions containing harmonics of the fundamental, as well as rational fractions known as sub- and ultraharmonics (Bader and Holland 2012; Hitchcock *et al.* 2011; O’Reilly and Hynynen 2012). An inertially collapsing bubble generates broadband acoustic emissions (Haworth *et al.* 2017). Histotripsy-induced bubble clouds produce harmonics (Bader *et al.* 2016a) and broadband (Canney *et al.* 2010; Vlaisavljevich *et al.* 2015b) emissions during the acoustic excitation. The presence of harmonics in the acoustic spectra may be indicative of geometric scattering of the non-linear histotripsy excitation by the bubble cloud (Maxwell *et al.* 2011b). After the acoustic excitation, shock waves are generated by the inertial collapse of the bubble cloud (Macoskey *et al.* 2018; Maxwell *et al.* 2013).

The primary means of detecting stable (Bader *et al.* 2015c; Sun *et al.* 2017) and inertial (Desjoux *et al.* 2013; Hockham *et al.* 2010) cavitation are single-element transducers. Backscattered emissions have been recorded with single-element transducers during intrinsic-threshold (Maxwell *et al.* 2013; Vlaisavljevich *et al.* 2015b), shock-scattering (Xu *et al.* 2005) and boiling (Canney *et al.* 2010) histotripsy excitations. A limitation of single-element transducers is the inability to triangulate bubble activity. Passive cavitation imaging (PCI) utilizes an ultrasound imaging array to detect and beam-form acoustic emissions generated by the mechanical oscillations of bubbles (Gyöngy *et al.* 2008; Salgaonkar *et al.* 2009). The resultant image maps cavitation activity spatially, and has been explored *in vitro* and *in vivo* to monitor thermal ablation with focused ultrasound (Arvanitis and McDannold 2013; Haworth *et al.* 2015; Jensen *et al.* 2012) and ultrasound-mediated drug delivery in real time (Choi *et al.* 2014; Coviello *et al.* 2015; Haworth *et al.* 2016; Kwan *et al.* 2015). Histotripsy bubble clouds have been monitored with PCI during liquefaction of a clot *in vitro* (Bader *et al.* 2016a). The spatial locations of acoustic power recorded with PCI and gray-scale recorded with plane wave B-mode imaging have been compared with the location of liquefaction in a prostate tissue phantom (Bader *et al.* 2018) (Fig. 15).

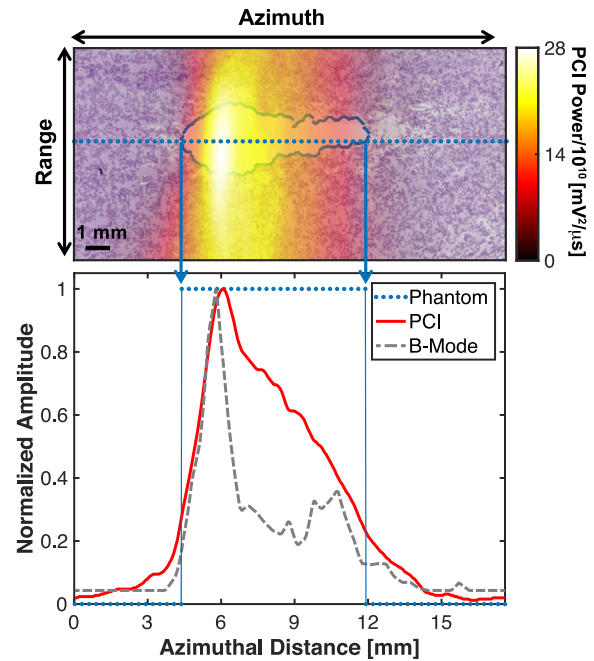


Fig. 15. Top: Passive cavitation image (PCI) registered with processed phantom image. The liquefaction zone border is outlined in blue. Bottom: Comparison of the PCI and plane wave B-mode image along the dotted line in the top panel and binary phantom image. For the phantom, values of 1 indicate liquefaction, and values of 0 indicate intact phantom. The histotripsy pulse (1-MHz center frequency, 10- μ s pulse duration, 18-MPa peak negative pressure) is propagating from left to right in the image (Bader *et al.* 2018). Reprinted with permission from *IEEE Transactions on Medical Imaging*, vol. 37, Bader *et al.*, Post hoc analysis of passive cavitation imaging for classification of histotripsy-induced liquefaction *in vitro*, p. 408. doi: 10.1109/TMI.2017.2735238. Copyright (2018). © IEEE.

PCI had a better area under the receiver operator characteristic curve, accuracy, sensitivity and negative predictive value for phantom liquefaction than plane wave B-mode imaging. However, B-mode and PCI were compared with the phantom only along the central axis of the histotripsy source (azimuth of the imaging array) because of the limited range resolution of the delay, sum and integrate beamformer used to process the passive images (Coviello *et al.* 2015). One means of improving the range resolution of PCI is to modify the geometry of the passive receiver array (Jones *et al.* 2018; O’Reilly *et al.* 2014). Elements within a hemispherical histotripsy source have been utilized to localize intrinsic threshold-induced cavitation, including quantifying the onset of bubble expansion and collapse (Macoskey *et al.* 2017). The limited bandwidth of the passive receivers prevented meaningful frequency analysis of the emissions (thereby the type of mechanical action), though this could be improved through the use of broadband receivers.

A limitation of the use of acoustic emissions to quantify bubble activity is the difficulty in system-independent cavitation emission measurements. Good agreement has been observed in the measured emission spectra and that predicted *via* the Yang/Church model for a single bubble (Collin and Coussios 2011). These results rely on careful calibration of the passive detector, with full knowledge of the spatial- and frequency-dependent sensitivity (Rich and Mast 2015a). The calibration accuracy of passive detectors is reduced in the presence of multiple emission sources characteristic of a histotripsy-generated bubble cloud (Rich and Mast 2015b). These calibration methods also assume spherical spreading of the emissions (Sboros et al. 2005), whereas scatter from histotripsy-generated bubble clouds has an angular dependence (Bader et al. 2018). The angular dependence of emissions may be imaged through the use of multi-element passive arrays (Haworth et al. 2017), but more work is still needed to quantify arbitrary cavitation emission source geometries.

Diagnostic ultrasound offers an improved temporal and spatial resolution compared with MRI at a reduced health care cost. But diagnostic ultrasound is inherently a 2-D imaging modality, cannot penetrate bony structures and lacks the soft-tissue contrast of MRI. Particularly for transcranial applications, MRI is important for histotripsy image guidance. Bubble clouds have been visualized with MRI (Allen et al. 2014, 2015). A sharp increase in the attenuation factor of gradient echo sequences was detected for insonation of an agar phantom between 24 and 28 MPa for intrinsic-threshold histotripsy pulses (Allen et al. 2014), indicating bubble cloud formation at peak negative pressures consistent with previous observations (Maxwell et al. 2013). These early findings indicate that the timing between the therapy pulse and the imaging gradients is critical for monitoring cavitation. In a follow-up study in *ex vivo* tissue, these sequences were tested against acoustic emissions recorded with a passive cavitation detector (Allen et al. 2015). Although there was agreement between the two methods on the cavitation threshold, the MR sequences were not as sensitive to bubble cloud formation compared with acoustic methods because of partial volume effects associated with MRI. These findings were based on 1-D MRI scans, but can be extended to additional dimensions.

In addition to volumetric oscillations, histotripsy-nucleated bubbles translate under primary radiation force in liquefied tissue (Xu et al. 2007). Monitoring of bubble cloud motion has been studied as a means to quantify tissue liquefaction. Color Doppler, which is sensitive to bubble motion (Ernst et al. 1996; Jauss and Zanette 2000), was found to be highly accurate for monitoring bubble cloud-induced fluid motion compared with

particle velocity imaging (Maxwell et al. 2014; Miller et al. 2016). Tissue motion around the bubble cloud was found to be chaotic immediately following insonation, after which coherent motion was observed. Movement was initially toward the source, but then reversed direction. The change in tissue direction may be due to a strong, symmetric collapse of the bubble cloud several milliseconds after the histotripsy pulse (Prieur et al. 2015; Xu et al. 2007). A significant correlation was observed between the time required for the coherent motion to reverse and liquefaction of a fibrin clot in *in vitro* (Zhang et al. 2015) and *ex vivo* porcine liver (Miller et al. 2016). Although these studies employed color Doppler (Szabo 2004), which is sensitive to the change in size of the bubble cloud post-excitation, additional information on the non-linear response of the bubble cloud can be extracted from pulse inversion Doppler (Li et al. 2014c). Tissue motion during treatment may result in artifacts of bubble-induced color Doppler and limit its accuracy during *in vivo* use. A combination of these techniques could be helpful in quantifying the bubble cloud activity, distribution of bubbles and degree of tissue liquefaction.

Quantify changes in the medium caused by bubble activity

Although monitoring of bubble activity is useful for image guidance, the clinical endpoint for histotripsy therapy is tissue liquefaction. To gauge treatment efficacy, it is critical to assess bubble dose-dependent changes in tissue structure. Ultrasound elastography has been investigated to assess the change in tissue stiffness over the course of histotripsy therapy. Tissue fractionation occurs over the application of tens to hundreds of histotripsy pulses. A strong decrease in tissue elasticity occurs as fractionation ensues (Hall et al. 2012) that can be visualized with acoustic radiation force impulse (ARFI) imaging (Palmeri and Nightingale 2011). Changes in the Young's modulus of *ex vivo* tissue exposed to histotripsy have been found to correlate with a decrease in the number of intact cellular nuclei (Wang et al. 2014). Furthermore, ARFI imaging is more sensitive to tissue fractionation than B-mode hypo-echogenicity. As the tissue liquefies, visualization of the shear waves generated in the ARFI sequence becomes difficult in the hypo-echoic focal zone. As such, the accuracy of elastography techniques for delineating the liquefaction zone may decrease during the final stages of the treatment.

In addition to diagnostic ultrasound modalities, histotripsy-induced liquefaction zones have been visualized with MRI at field strengths between 7 and 9 T (Allen et al. 2014, 2015; Khokhlova et al. 2009; Vlasisavljevich et al. 2016c). Changes in the structure of media have been clearly delineated with T2-, T1- and diffusion-weighted

and contrast-enhanced sequences *in vitro* and *in vivo* (Allen *et al.* 2017; Smollock *et al.* 2018). Lesion appearance varies based on tissue viability (*in vivo* vs. *ex vivo*) and the time at which the image is acquired relative to the histotripsy therapy. The MR pixel intensity within the focal zone increases immediately post-liquefaction of *ex vivo* tissue (Allen *et al.* 2015; Kim *et al.* 2013; Vlasisavljević *et al.* 2013b), but decreases immediately after *in vivo* insonation (Vlasisavljević *et al.* 2016c). Over the first few days post-treatment, the MR pixel intensity of the treatment zone increases relative to the background, suggesting iron-based blood products and edema are present within the lesion. The pixel intensity gradually decreases over time and appears to be resolved within 28 d sans a thin rim (Vlasisavljević *et al.* 2016c), indicating the accumulation of deoxygenated red blood cells (Kim *et al.* 2013). It was noted that histotripsy ablation zones could be clearly visualized immediately post-insonation with T2-weighted imaging. Thermally generated ablation zones can require several weeks to be fully visible under T2 imaging, in contrast to histotripsy-generated lesions (Coad *et al.* 2003; Marrero and Pelletier 2006).

CONCLUSIONS

Over the course of this review, the salient features of histotripsy bubble activity were distinguished from other forms of cavitation. Although seemingly disparate insonation schemes are utilized in histotripsy, there remain unifying mechanisms and concepts that govern this specific type of bubble behavior. In particular, supplying sufficient tension to activate expansion of a ubiquitous population of nuclei can produce consistent formation of bubbles in tissue. Intrinsic-threshold and shock-scattering histotripsy produces tension that exceeds an intrinsic cavitation threshold of the medium under physiologic conditions. Experiments indicate that boiling histotripsy heats the medium rapidly, reducing the intrinsic threshold to a level near the incident peak negative pressure. Evidence that multiple types of nucleation can occur simultaneously (*e.g.*, shock-scattering and boiling [Khokhlova *et al.* 2011]) confound complete discretization of histotripsy nucleation schemes.

The expansion of the nanoscale nuclei to bubbles tens of micrometers to millimeters serves as the acting force of histotripsy therapy, imparting sufficient strain to lyse the surrounding cells and break up the extracellular components (Vlasisavljević *et al.* 2016d). However, there still remains a gap in knowledge of the fractionation process. Qualitative correlations exist between bubble activity and bio-effects, but quantification of these relationships is still lacking. Strain induced by bubble volumetric oscillations has been reported to induce cell lysis (Vlasisavljević *et al.* 2016d). Jetting,

translation and atomization have also been found to occur during the histotripsy cavitation process. Which of these mechanisms are causal to tissue liquefaction and which are superfluous bubble activity (*e.g.* changes in bubble activity due to changes in the environment) remain unknown. The highly non-linear action of cavitation also hampers straightforward identification of the appropriate parameter set for promotion of a specific bubble activity. There is no singular “knob” to adjust to engage bubble jetting, for instance. The interaction of histotripsy

bubbles with the insonation source and their surrounding medium, as well as promotion of specific bubble activities associated with an intended bio-effect, should remain active areas of research. Finally, image guidance methods are needed for feedback of the intended histotripsy-induced bubble activity. Despite these gaps in knowledge, the continual progress being made toward understanding bubble behavior in histotripsy has already revealed several reliable methods for tissue ablation with the potential to replace many surgical procedures in the future.

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